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Fall26 Modern Software Concepts

1. How many entries in your database are from applicants who applied for Fall 2026?

Answer: 33207

SQL: SELECT COUNT(*) AS fall_2026_entries
FROM applicants
WHERE term = %s;

Explanation: SELECTCOUNT(*) gets the number of rows in the database. From specifies the database, and WHERE term = %s is the placeholder for the category term which AS fall_2026_entries highlights.

2. Among entries that provide a nationality classification, what percentage are international students?

Answer: 48.48

SQL: SELECT ROUND(100.0 * COUNT(*) FILTER (WHERE us_or_international = 'International') /
NULLIF(COUNT(*), 0),2) AS international_percentage FROM applicants WHERE us_or_international IS
NOT NULL;

Explanation: SELECT is explained above, it picks out values. The ROUND is for the required 2 decimal place. The (100 * FILTER COUNT/COUNT) is to get the percentage. WHERE specifies for internationals for the term “us_or_international”. The NULLIF is to prevent division by zero or something wonky. The FROM applicants WHERE us_or_international NOT NULL is to not evaluate unknown cases.

3. What are the average GPA, GRE Quantitative, GRE Verbal, and GRE Analytical Writing scores of applicants who provide each metric?

Answer: average gpa = 3.77, average gre = 260.93, average gre v = 161.49, average gre writing = 9.06

SQL: SELECT ROUND(AVG(gpa)::NUMERIC, 2) AS average_gpa, ROUND(AVG(gre)::NUMERIC, 2) AS
average_gre, ROUND(AVG(gre_v)::NUMERIC, 2) AS average_gre_verbal,
ROUND(AVG(gre_aw)::NUMERIC,2) AS average_gre_writing FROM applicants;

Explanation: SELECT the ROUND average by getting AVERAGE(gpa) column with only the numbers that are filled in by the applicants. The ::NUMERIC portion converts the result of average into PostgreSQL's numeric data type. Which can be rounded.

4. What is the average GPA of American applicants who applied for Fall 2026?

Answer: 3.79

SQL: SELECT ROUND(AVG(gpa)::NUMERIC,2) AS average_american_gpa FROM applicants WHERE term
= %s AND us_or_international = %s AND gpa IS NOT NULL;

Explanation: The logic is the same as the one above. The only difference is that us_or_international field is specified with the run_query(cursor, question, SQL, <specification>) part where term is Fall 2026 and us_or_international is American and the fields where GPA is not expressly written are casted out with IS NOT NULL.

5. What percentage of Fall 2025 entries are acceptances?

Answer: 39.45

SQL: SELECT ROUND(100.0 * COUNT(*) FILTER (WHERE status = 'Accepted') / NULLIF(COUNT(*), 0),2) AS fall_2025_acceptance_percentage FROM applicants WHERE term = %s;

Explanation: The logic is the same as answering question number 2. The difference is that this time status 'Accepted' are the ones selected. The filter in WHERE being Fall 2025 is specified in the run_query() method like in the previous question.

6. What is the average GPA of accepted applicants who applied for Fall 2026?

Answer: 3.78

SQL: SELECT ROUND(AVG(gpa)::NUMERIC, 2) AS accepted_average_gpa FROM applicants WHERE term = %s AND status = %s AND gpa IS NOT NULL;

Explanation: The logic is the same as question 3 above but now WHERE has three filters where the gpa category is filled and the term and status are both being specified in the run_query like above—in this case they are Fall 2026 for term and accepted for status.

7. How many entries are from applicants who applied to Johns Hopkins University for a master's degree in Computer Science?

Answer: 19

SQL: SELECT COUNT(*) AS johns_hopkins_cs_masters_entries FROM applicants WHERE LOWER(llm_generated_university) = LOWER(%s) AND LOWER(llm_generated_program) = LOWER(%s) AND degree ILIKE %s;

Explanation: SELECT COUNT(*) gets the number of applicants and use the lower case of everything in lower(llm_generated_university) = lower(johns hopkins) and get the same for the program. As for the degree, using ILIKE to get case-insensitive versions of “master”.

8. How many Fall 2026 entries are acceptances from applicants applying for a PhD in Computer Science at one of the following universities?

Answer: raw text count = 30

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SQL: WITH raw_text_matches AS (SELECT COUNT(*) AS match_count
FROM applicants AS applicant
WHERE applicant.term = %s
AND applicant.degree = %s
AND applicant.status = %s
AND POSITION( 'computer science' IN LOWER(applicant.program)) > 0
AND EXISTS (SELECT 1 FROM UNNEST(%s::TEXT[]) AS selected(university)
WHERE POSITION(LOWER(selected.university) IN LOWER(applicant.program)) > 0)),
llm_matches AS (SELECT COUNT(*) AS match_count
FROM applicants
WHERE term = %s
AND degree = %s
AND status = %s
AND LOWER(llm_generated_program) = 'computer science'
AND LOWER(llm_generated_university) = ANY(%s::TEXT[]))
SELECT raw_text_matches.match_count AS raw_text_count,
llm_matches.match_count AS llm_standardized_count,
llm_matches.match_count - raw_text_matches.match_count AS count_difference
FROM raw_text_matches CROSS JOIN llm_matches;
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Explanation: This formation allows for question number 8 and 9 to be answered simultaneously. We want raw text count for Question 8. This will be done by creating a match_count for fall 2026, acceptances, and specifically for PhD in computer science and the selected universities, which there is a special list for. I use select count(*) gets the number of row. Database is specified with applicants and the filter is done with term, degree, and status with the run_query() method with Fall 2026, PhD, and accepted respectively for each of the fields. The first position function finds “computer science” in program. Using unnest, the selected universities list is separated into strings and the exists part checks to see if the program is the selected universities with the second position function, the select 1 determines the true/false.

9. How many Fall 2026 entries are acceptances from applicants applying for a PhD in Computer Science at one of the following universities? [BUT WITH LLM FIELDS]

Answer: llm count = 15, hence with original 30, the difference is 30 – 15, which is 15

SQL: Same as Question 8's

Explanation: To get the difference, the exact same query but with llm generated fields for program and university used to specify—this will be categorized under llm_matches. Afterwards, just report the number and differences— the algebra is done by combining the two together with cross join.

10. What are the international student percentage among accepted NYU master's applicants?

Answer: 45.49

SQL: SELECT ROUND(100.0 * COUNT(*) FILTER (WHERE us_or_international = 'International') /
NULLIF(COUNT(*), 0),2) AS nyu_international_percentage
FROM applicants

WHERE status = %s

AND LOWER(llm_generated_university) = LOWER(%s)

AND degree ILIKE %s;

Explanation: Same logic as question 2. I use llm generated field to filter for NYU and run_query() to specify status and degree.

11. What is the average GPA of students accepted for Johns Hopkins masters?

Answer: 3.68

SQL: SELECT ROUND(AVG(gpa)::NUMERIC, 2) AS johns_hopkins_average_gpa
FROM applicants WHERE status = %s
AND LOWER(llm_generated_university) = LOWER(%s)
AND degree ILIKE %s AND gpa IS NOT NULL;

Explanation: Same logic as question 3. I use llm generated field here too and the average is copied from the above as well.